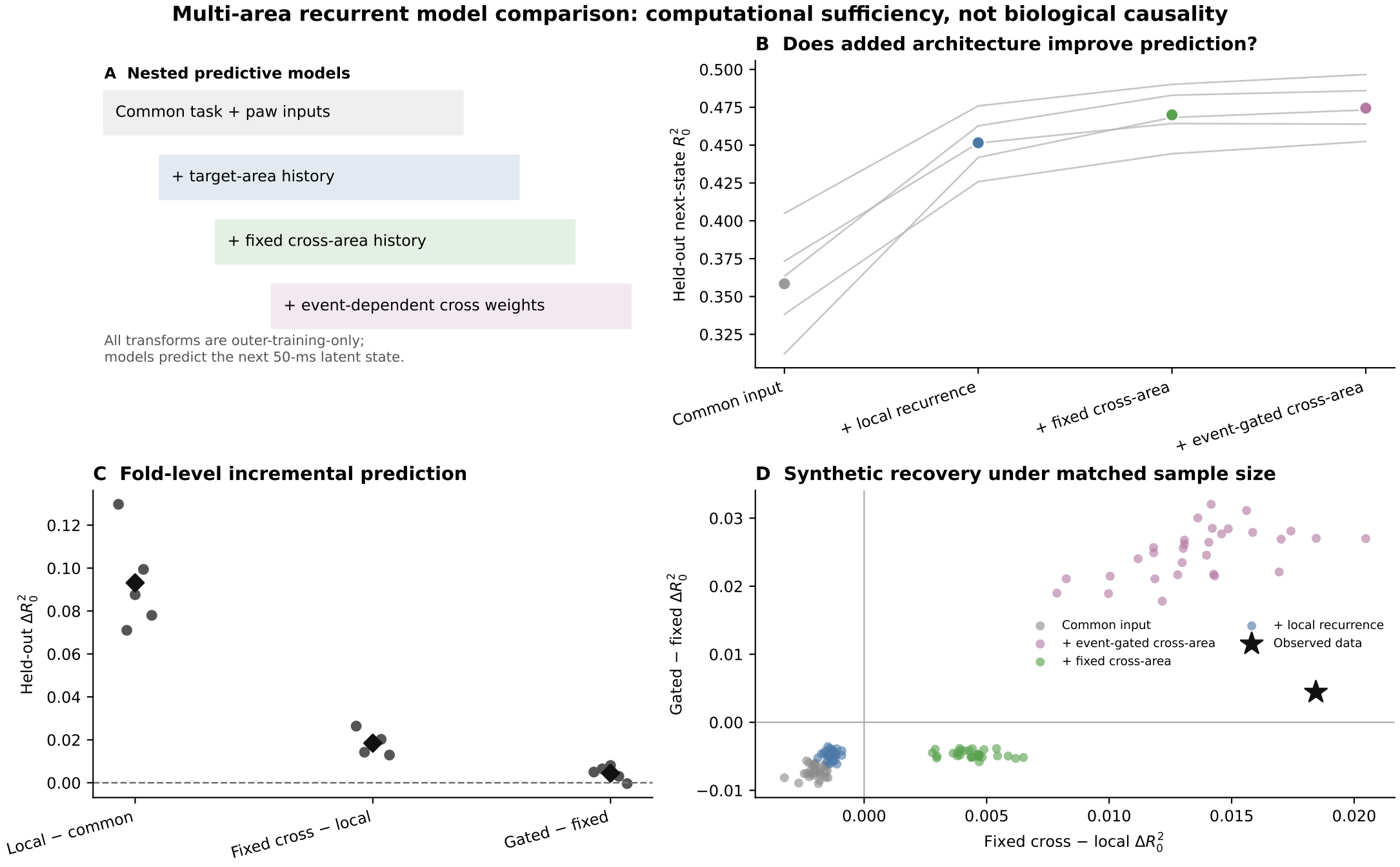
Multi-area recurrent model comparison: computational sufficiency, not biological causality
A Nested predictive models
Common task + paw inputs
+ target-area history
+ fixed cross-area history
+ event-dependent cross weights
All transforms are outer-training-only;
models predict the next 50-ms latent state.
B Does added architecture improve prediction?
Held-out next-state R_0^2
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0.325
Common input
+ local recurrence
+ fixed cross-area
+ event-gated cross-area
C Fold-level incremental prediction
Held-out ΔR_0^2
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Local − common
Fixed cross − local
Gated − fixed
D Synthetic recovery under matched sample size
Gated − fixed ΔR_0^2
0.03
0.02
0.01
0.00
−0.01
0.000
0.005
0.010
0.015
0.020
Fixed cross − local ΔR_0^2
Common input
+ event-gated cross-area
+ fixed cross-area
+ local recurrence
Observed data